

UNIVERSITY OF DELAWARE · DEPARTMENT OF COMPUTER & INFORMATION SCIENCES

DAS — Initiative for Dependable Agentic Systems

Agentic systems that safely and reliably automate complex scientific and operational work



Dong Dai

Co-Director



Yixiang Deng

Co-Director



Weisong Shi

Senior Advisor



Xing Gao

Faculty Member

Team — Execution Evidence



Dong Dai

Associate Professor · Co-Director

IOAgent — IPDPS'25

STELLAR — SC'25

Agentic optimization for HPC storage and I/O workflows.



Yixiang Deng

Assistant Professor · Co-Director

Nature Communications

Cell Reports

npj Digital Medicine

AI-for-science for discovery and digital health.



Weisong Shi

Department Chair · Senior Advisor

SENIOR ADVISOR

Edge / Physical AI Strategy

Advises strategy, industry engagement, and embodied-agent directions.



Xing Gao

Assistant Professor · Faculty Member

MCP security — DSN'26

USENIX Sec'24 · CCS/S&P'24

Security for agent protocols, cloud, and supply chains.

DAS Mission - Three Pillars



Student Training

Courses · capstones · graduate research



Industry Collaboration

Consulting · TSC · pilot projects



Research Collaboration

Agent systems × domain science

Research Areas

DONG DAI

Agent Systems for System Optimization

Agents that diagnose, tune, and explain complex computing systems.

Representative work

IOAgent — IPDPS'25

STELLAR — SC'25

ION — HotStorage'24

XING GAO

Security for LLM & Agent Systems

Security analysis for agent protocols, cloud infrastructure, and software supply chains.

Representative work

MCP security — DSN'26

GPU cache attacks —

USENIX Sec'24

CI/CD security — CCS'24 / S&P'24

YIXIANG DENG

Agent × Science and Health

Scientific AI for biological discovery, digital health, and closed-loop decision support.

Representative work

AI pancreas — bioRxiv

Glucose prediction — npj

Digital Med

Immune profiling — Nature Comms / Cell Reports

SENIOR ADVISOR

Expansion Area: Physical Agents

Open area for collaborators working on agents connected to instruments, robots, edge devices, or vehicles.

Status

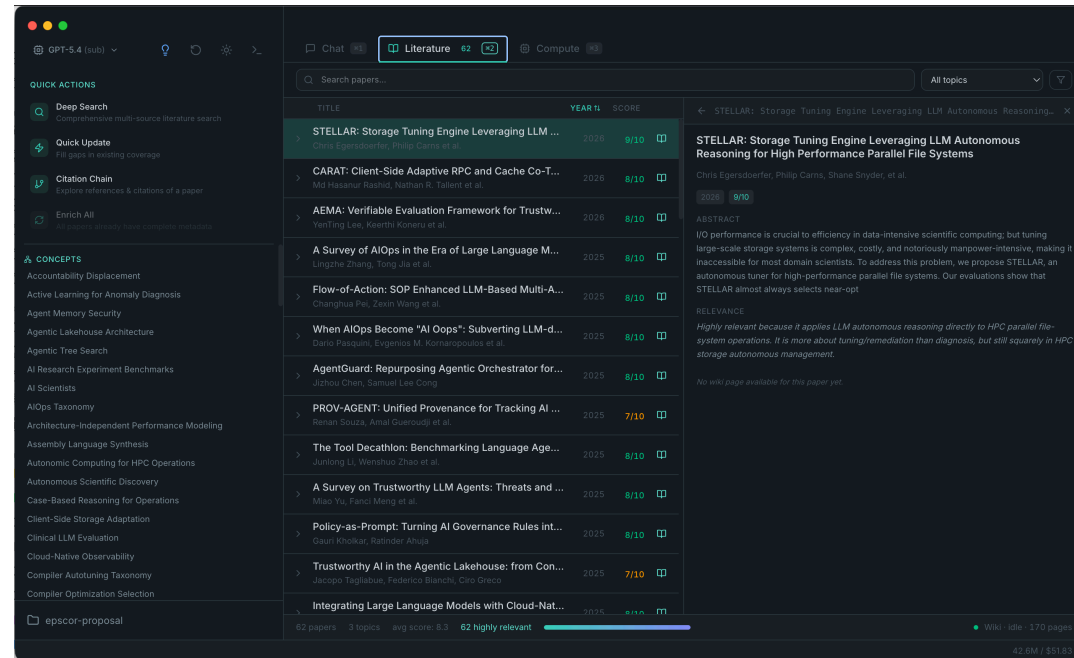
Advisor-guided direction

Seeking faculty affiliates

Seeking industry pilots

Example Product — Research Copilot

Agentic Systems Designed for Scientific Research



Research Copilot · accountable research coworkers ·
daidong.github.io/PiPilot

COWORK WITH CLAUDE

Literature · data analysis · writing ·
project memory

Provenance / Audit

Trace every action and source.

Local Knowledge

Paper wiki and memory stay local.

Local + Cloud Models

Claude/GPT now; local models when needed.

How We Work with Industry

Collaboration can start as a service contract and deepen into funded research.

①

TSC / Consulting

Contracted expert help through UD's Technology Service Center.

②

Pilot Research

A 3–6 month project around one concrete workflow and deliverable.

③

Gift / Project Funding

Support students and exploratory topics without over-scoping too early.

④

Strategic Partnership

Shared roadmap, repeat projects, student pipeline, and joint outputs.

Start small → de-risk a pilot → fund a focused project → build a durable research channel.

TSC: sites.udel.edu/technology-service-center

What We Are Looking For

Help us choose the first dependable-agent problems worth building.



Founding Advisor

name / organization

TECHNICAL JUDGMENT

doors

use cases

validation

Pilot Problems

one real workflow · one clear success criterion

Flexible Support

gift · project · student support

The first advisor slots should pressure-test the roadmap and open the right conversations.

LET'S BUILD SOMETHING TOGETHER

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Scan to visit our site

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